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What Are All the Things Existing Computable Knowledge Can Compute?

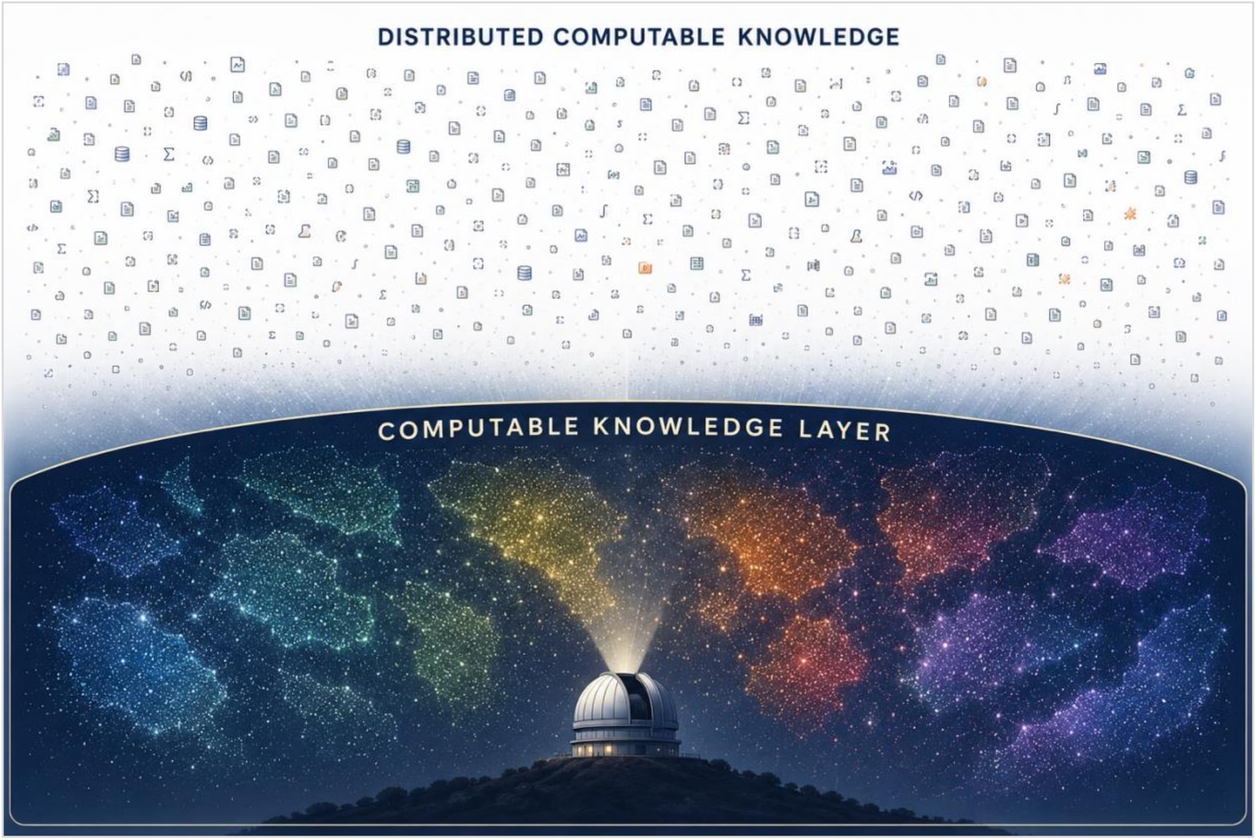

Figure. *The Challenge of Revealing What Can Be Computed.* In this figure, distributed computable knowledge resources collectively support a vast space of potential computations. Although the resources themselves are visible, the computational capabilities they enable are often not. The Computable Knowledge Layer is conceptualized as an infrastructure that makes this latent capability landscape discoverable, helping answer a fundamental question: Within a large universe of computable knowledge, what can be computed?

INTRODUCTION: THE COMPUTABLE KNOWLEDGE LAYER IS MISSING

Modern computing infrastructure can readily answer questions about data, documents, services, and software with support from metadata. Search engines identify documents. Repositories organize software. Registries list services. Along these lines, a new need to discover what existing computable knowledge can compute is becoming increasingly relevant.

Despite decades of progress in information infrastructure, there is no broadly adopted mechanism for figuring this out. Computable knowledge is widely distributed but not organized and described in ways that enable immediate discovery of its computational capabilities. This is a problem because many computational activities depend not merely on data or software, but on computable knowledge, e.g., predictive models or decision rules. Each artifact may support computation locally or within particular systems, but its contribution to a larger set of capabilities is rarely visible. As computable knowledge accumulates, the challenge shifts from understanding individual artifacts to discovering the computational capabilities collectively present within large bodies of computable knowledge, bodies that could be aggregated into a massive computable knowledge layer or layers.

The World Wide Web addressed a similar problem for information resources. Its transformative contribution was not the creation of documents, but their tagging, indexing, and subsequent discovery on a huge scale. Computable knowledge presents a related but distinct challenge. The objective is to describe, index, and align computable knowledge artifacts so that it becomes easy to discover how they may contribute to realizing computational goals. Viewed from this perspective, modern computing possesses mature infrastructures predicated on metadata for data, documents, software, services, and increasingly for content generation. What remains comparatively underdeveloped is corresponding infrastructure for making computational capabilities discoverable on an equally large scale. Such discovery requires new mechanisms for consistently relating computational goals to available computable knowledge.

HERE, THERE, AND EVERYWHERE: COMPUTABLE KNOWLEDGE ALREADY EXISTS

Computational capabilities predicated on computable knowledge are already distributed throughout the global digital ecosystem. Existing systems provide limited support for understanding what may be computable given the computable knowledge currently scattered about.

This challenge is not new. Decades before the emergence of generative AI, the Digital Object architecture proposed an infrastructure in which digital resources could be reliably identified, quickly discovered, and then acted upon through their metadata. In this architecture, a strong emphasis on persistent referenceability is arguably as important as the metadata themselves. Before resources can be described, aligned, or combined, they must first be identifiable as distinct things to which stable references can be made. Many concepts now associated with modern digital infrastructure—including persistent identifiers, typed objects, machine-readable metadata, indexing and resolution services—were present in early Digital Object architecture.

Yet despite the call to make it so, the Digital Object architecture never became a dominant organizing principle of the Web. It may have arrived before the conditions required for its widespread adoption. Metadata creation and maintenance were expensive, while relatively few machine consumers existed to exploit stable object references and rich metadata at scale.

Those conditions are now changing. AI systems are reducing the cost of metadata generation while emerging as large-scale consumers of external resources, including computable knowledge. AI-enabled systems and agents, and the software tools they use, increasingly depend upon the ability to identify, characterize, and apply computable knowledge at scale.

It is unclear whether the combination of more mature Digital Object concepts, vast stores of computable knowledge, formalized metadata semantics, and AI-based machine consumers establish sufficient conditions for the new layer to emerge. If so, then a new global Computable Knowledge Layer added to the Web may make computational capabilities easily and widely discoverable. Its significance lies not in creating new computable knowledge, but in making visible all the computations that the world's computable knowledge can collectively generate. That visibility arises not from the computable knowledge itself, but from metadata about it.

A Computable Knowledge Layer seems more likely to emerge through incremental accumulation rather than through large-scale construction efforts. Computable knowledge, metadata, and machine consumers already exist at scale and continue to grow through largely independent activity. The work is not so much to create a new layer of computable knowledge, rather it is to establish mechanisms that reveal latent computational capabilities in a layer that is under formation already. Like many important capabilities of the Web itself, a more coherent layer may emerge from independently created resources that get described and connected using only lightweight conventions.

PERSPECTIVE SHIFT: FROM INDIVIDUAL ARTIFACTS TO LAYERS WITH LATENT CAPABILITIES

Documents, datasets, software applications, and computable knowledge artifacts share some characteristics but also differ in important ways. All are subject to questions of authority, quality, applicability, and generalizability. Such assessments are inherently contextual, reflecting the perspectives, objectives, and standards of individuals and communities rather than fixed properties of the artifacts themselves. Unlike documents that communicate information, datasets that represent observations, and applications that facilitate operations, computable knowledge artifacts support evidence-based computation. Their importance therefore extends beyond their informational content to the computational capabilities they may contribute toward realizing particular goals.

This distinction is fundamental. Computable knowledge artifacts are observable objects. Computational capabilities are not. A predictive model, decision rule, classifier, workflow, or calculator may be directly inspected, described, referenced, and exchanged. The computational capabilities such artifacts support, particularly when considered collectively, are often far less apparent. The evident organizing need addressed by a Computable Knowledge Layer is therefore not the discovery of artifacts themselves, but the discovery of the computational capabilities they individually and collectively make possible.

A shift in perspectives is in order from the artifacts themselves to the computational capabilities they support when aggregated into one or more computable knowledge layers. Viewed through the lens of individual artifacts, these capabilities appear straightforward. Viewed collectively, they become considerably more difficult to characterize. Large layers of computable knowledge are new scientific objects of study that may ultimately support thousands, millions, or potentially far larger numbers of computational possibilities. Many of these capabilities may overlap, complement one another, or combine in ways that are not immediately apparent from inspection of individual artifacts. Revealing such relationships requires more than metadata alone. It requires the ability to refer consistently to inputs, outputs, concepts, entities, and computable knowledge artifacts themselves. Without shared references, computational relationships remain difficult to describe and discover, and layers can never fully form.

The most critical challenge is to understand the computational capabilities present within a body of computable knowledge and determine which capabilities may contribute to achievement of any computational goal. This broader perspective sees computational capability itself as an infrastructure concern. Importantly, the capability is not created by any new layer. The capability already largely exists within the available computable knowledge. The role of the computable knowledge layer is to make that capability visible once computable knowledge metadata become more available. This kind of visibility is a precursor to realizing computational goals on demand.

WHAT THE NEW LAYER PROVIDES: EMERGING PROFILES OF COMPUTATIONAL CAPABILITY

Imagine being handed access to every predictive model, decision rule, classifier, workflow, calculator, and computational procedure currently available on the Internet.

What could you compute?

Surprisingly, this question is difficult to answer. When large numbers of such artifacts are considered together, they acquire a kind of collective capability profile. Some computations become possible only through combinations of available computable knowledge. Others never do. Determining which capabilities are present requires interrogating the layer itself.

The real challenge is to discover computational capability itself, and for that, artifact metadata are key. The desired computable knowledge layer is formed with a mix of computable knowledge and machine-actionable metadata that are rich enough to make what may be computed observable throughout the entire layer, thereby allowing its computational capabilities to begin to be profiled.

COMPLETING THE PICTURE: NEW ROLES FOR EXPLICIT GOALS

People rarely seek computable knowledge for its own sake. They seek it because they wish to accomplish something. A clinician wishes to estimate risk. A scientist wishes to predict an outcome. An engineer wishes to compute a physical limit. A researcher wishes to classify, simulate, infer, recommend, estimate, or explain something using computation.

These objectives may all be viewed as *computational goals*.

The introduction of such goals fundamentally changes the computational capability discovery problem. Without a goal, distinct bodies of computable knowledge inside the computable knowledge layer may appear as a large collection of artifacts and capabilities. With a goal, the same collection can be examined in terms of its relevance to realizing a desired computation.

This distinction mirrors an important development in the history of the Web. Documents became broadly discoverable only after information needs could be expressed and matched against available resources using keywords. Computational goals play a similar role for computable knowledge. Like keywords, they inform underlying infrastructural mechanisms through which potentially relevant computational capabilities can be identified. In this manner, computational goals do more than specify desired computations. They also function as probes through which the computational capabilities of a layer can be explored.

Adding explicit goals subtly shifts the central question. The challenge is no longer simply to understand what computational capabilities exist. The challenge becomes determining which capabilities may contribute to realization of explicit computational goals. This transition motivates a new form of discovery in which computational goals are used to reveal potentially relevant computational capabilities within a vast computable knowledge layer. This new process leverages discovery but is more accurately characterized as a process of computational goal and computable knowledge alignment.

Importantly, this alignment process does not determine which computational possibilities are best, correct, authoritative, or appropriate. Assessments of quality like these are made by individuals and communities in relation to goals, contexts, and available evidence. Such assessments come about through observation, experience, and the continued evaluation of outcomes arising from the downstream application of computable knowledge. Alignment simply surfaces possibilities that may contribute to realization of a computational goal. Judgments about their suitability remain the responsibility of the goal-setter or the communities they represent. In this way, computational goals serve not only as mechanisms for capability discovery, but also as part of the context within which such judgments are made.

GOAL-DRIVEN ALIGNMENT: REVEALING RELEVANT CAPABILITY ONE GOAL AT A TIME

The introduction of computational goals establishes a fundamental role for the Computable Knowledge Layer. Explicit computational goals provide the organizing principle that makes computational capability discovery possible on a goal-relevant basis. A computational goal expresses a desired computation. Examples include estimating a risk, classifying an entity, predicting an outcome, determining a quantity of interest, selecting among alternatives, or generating an explanation. Once a goal is expressed, the challenge becomes identifying computable knowledge that may contribute to its realization.

This process of alignment is conceptually related to search but differs in an important respect. Search begins with an information need expressed in keywords and seeks relevant documents. Alignment begins with a computational goal and seeks computable knowledge capable of contributing to its realization. Doing so requires mechanisms for relating desired outputs to available computational capabilities and, ultimately, for relating outputs produced by some forms of computable knowledge to inputs required by others.

Importantly, alignment reveals more than individual artifacts. Each alignment exercise also exposes information about the layer itself. A successful alignment demonstrates that the layer contains computable knowledge capable of contributing to realization of an explicit goal. An unsuccessful alignment indicates that relevant capabilities are absent from the layer.

Viewed collectively, repeated alignment activities begin to reveal the computational capability profile of the entire layer. Different goals probe different portions of the available computable knowledge. Over time, these observations provide evidence regarding what the layer can and cannot compute.

This perspective suggests that alignment serves two purposes simultaneously. It identifies computable knowledge relevant to a specific computational goal. It also functions as a mechanism for revealing many of the computational capabilities collectively present within the layer.

In this sense, alignment may be understood as a process for making computational capability observable. The resulting capability profile that gets revealed over time is not created by the layer. It already exists within it. Alignment simply makes that capability increasingly visible, one computational goal at a time. Whether alignment, conceived of in this way, proves effective under real-world conditions is an open area of investigation.

CONCLUSION: CONSIDER THE CAPABILITIES OF A COMPUTABLE KNOWLEDGE LAYER AT SCALE

The world's computable knowledge already exists at large scale. The emergence of computable knowledge as a distinct class of scientific artifact, combined with advances in metadata technologies, Digital Object concepts, and AI-enabled machine consumers, suggests that the conditions may now exist for one or more very large computable knowledge layers to emerge. Making computational goals explicit offers an approach through which the needs, values, constraints, and judgments of goal-setters can inform discovery-dependent alignment within computable knowledge layers. The significance of these layers lies not in creating new computable knowledge. It lies in answering a pressing question of our age that has astonishing scope and breadth. What are all the things existing computable knowledge can compute?